

Supplementary Information: Autonomous ADC Linker Design

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1 Full context ranking (all 24 contexts)

Handle	Trigger	mean	best	SD	seeds
Oxyamine	Non-cleavable	0.84	0.85	0.01	3
Maleimide	Non-cleavable	0.83	0.84	0.01	3
Bromoacetamide	Non-cleavable	0.82	0.83	0.01	3
NHS_ester	Non-cleavable	0.81	0.81	0.00	3
Disulfide	Non-cleavable	0.75	0.78	0.03	3
Maleimide	Val-Ala-PABC	0.76	0.77	0.02	3
Oxyamine	Val-Ala-PABC	0.75	0.76	0.01	3
NHS_ester	Val-Ala-PABC	0.75	0.76	0.01	3
Bromoacetamide	Val-Ala-PABC	0.74	0.74	0.00	3
DBCO	Non-cleavable	0.69	0.73	0.03	3
Maleimide	Val-Cit-PABC	0.69	0.71	0.03	3
Disulfide	Val-Ala-PABC	0.68	0.70	0.01	3
Bromoacetamide	Val-Cit-PABC	0.66	0.68	0.02	3
Oxyamine	Val-Cit-PABC	0.66	0.67	0.01	3
NHS_ester	Val-Cit-PABC	0.65	0.67	0.02	3
DBCO	Val-Ala-PABC	0.61	0.65	0.03	3
Disulfide	Val-Cit-PABC	0.60	0.63	0.02	3
DBCO	Val-Cit-PABC	0.52	0.53	0.01	3
Bromoacetamide	Glucuronide	0.08	0.09	0.00	3
Oxyamine	Glucuronide	0.08	0.08	0.00	3
Maleimide	Glucuronide	0.08	0.08	0.00	3
NHS_ester	Glucuronide	0.08	0.08	0.00	3
Disulfide	Glucuronide	0.08	0.08	0.00	3
DBCO	Glucuronide	0.06	0.06	0.00	3

2 Commercial linker benchmark

Name	ADC / class	composite	QED	MW
mc-Val-Cit-PABC	brentuximab vedotin (Ad...	0.52	0.116	573
mc-Val-Ala-PABC	Val-Ala class	0.59	0.244	487
mc-GGFG	trastuzumab deruxtecan ...	0.58	0.128	530
MCC (non-cleavable)	trastuzumab emtansine (...)	0.67	0.737	237
SPDB disulfide	mirvetuximab soravtansi...	0.74	0.600	229
sulfo-SPDB	hindered disulfide class	0.76	0.444	309
AcBut hydrazone	gemtuzumab ozogamicin (...)	0.00	0.340	236
beta-glucuronide	glucuronide linker class	0.00	0.156	508
maleimidocaproyl (mc)	mc handle	0.68	0.514	211

3 Table S2: Top designed linkers (full SMILES)

Context	Score	SMILES
Oxyami/Non-cl	0.85	<chem>NOCC(=O)n1cc(C(=O)NCc2ccccc2)c(=O)[nH]c1=O</chem>
Maleim/Non-cl	0.84	<chem>Cn1c(=O)n(CC(=O)NCc2ccccc2)c(=O)n(CC(=O)N2C(=O)C=CC2=O)c1=O</chem>
Maleim/Non-cl	0.84	<chem>O=C(Cn1c(=O)[nH]c(=O)n(CCN2C(=O)C=CC2=O)c1=O)NCc1ccccc1</chem>
Maleim/Non-cl	0.84	<chem>O=C(Cn1c(=O)[nH]c(=O)n(CC(=O)N2C(=O)C=CC2=O)c1=O)NCc1ccccc1</chem>
Oxyami/Non-cl	0.84	<chem>Cn1c(=O)n(CCON)c(=O)c2c1c(=O)n(CC(=O)NCc1ccccc1)c(=O)n2C</chem>
Oxyami/Non-cl	0.84	<chem>Cn1c(=O)c2c(=O)n(CCON)c(=O)[nH]c2c(=O)n(CC(=O)NCc2ccccc2)c1=O</chem>
Oxyami/Non-cl	0.83	<chem>Cn1c(=O)c2c(ncn2CCON)c(=O)n(CC(=O)NCc2ccccc2)c1=O</chem>
Oxyami/Non-cl	0.83	<chem>CN(C(=O)NCc1ccccc1)c1c(N)n(CCON)c(=O)[nH]c1=O</chem>
Maleim/Non-cl	0.83	<chem>Cn1cc(NC(=O)CN2C(=O)C=CC2=O)c(=O)n(CC(=O)NCc2ccccc2)c1=O</chem>
Bromoa/Non-cl	0.83	<chem>O=C(Cn1c(=O)nc(NC(=O)NC(=O)CBr)[nH]c1=O)NCc1ccccc1</chem>
Bromoa/Non-cl	0.83	<chem>O=C(CBr)NS(=O)(=O)NC(=O)NC(=O)NCc1ccccc1</chem>
Oxyami/Non-cl	0.83	<chem>CC(=O)NC(=O)c1c(C(=O)NCc2ccccc2)[nH]c(=O)n(CCON)c1=O</chem>
Oxyami/Non-cl	0.83	<chem>NOCCn1c(=O)c2cc(C(=O)NCc3ccccc3)ccc2n(C(N)=O)c1=O</chem>
Oxyami/Non-cl	0.83	<chem>NOCCCNC(=O)c1cn(C(=O)NCc2ccccc2)c(=O)[nH]c1=O</chem>
Maleim/Non-cl	0.83	<chem>O=C(CN1C(=O)C=CC1=O)Nc1n[nH]c(C(=O)NCc2ccccc2)n1</chem>
Bromoa/Non-cl	0.83	<chem>O=C(CBr)NC(=O)Cn1nc(NC(=O)NCc2ccccc2)c(=O)[nH]c1=O</chem>
Oxyami/Non-cl	0.83	<chem>NOCC(=O)Nc1c(C(=O)NCc2ccccc2)[nH][nH]c1=O</chem>
Oxyami/Non-cl	0.83	<chem>NOCCn1c(=O)c2cc(C(=O)NCc3ccccc3)ccc2n(CC(N)=O)c1=O</chem>
Oxyami/Non-cl	0.83	<chem>NOCCn1c(N)nc(N2CCN(C(=O)NCc3ccccc3)CC2)cc1=O</chem>
Maleim/Non-cl	0.82	<chem>O=C(Cn1c(=O)c(=O)n(CC(=O)N2C(=O)C=CC2=O)c2ccccc2c1=O)NCc1ccccc1</chem>
Maleim/Non-cl	0.82	<chem>Nn1c(SCC(=O)N2C(=O)C=CC2=O)nnc(C(=O)NCc2ccccc2)c1=O</chem>
Maleim/Non-cl	0.82	<chem>O=C(NCc1ccccc1)c1cc(C(=O)NNC(=O)N2C(=O)C=CC2=O)[nH]n1</chem>
Oxyami/Non-cl	0.82	<chem>NOCCn1c(=O)c(C(=O)NCc2ccccc2)cc2cnc(N)nc21</chem>
Bromoa/Non-cl	0.82	<chem>O=C(Cn1cnc2c(nnn2CC(=O)NC(=O)CBr)c1=O)NCc1ccccc1</chem>
Oxyami/Non-cl	0.82	<chem>NOCCC(=O)c1c(C(=O)NCc2ccccc2)[nH]c(=O)[nH]c1=O</chem>

4 Table S3: Boltz-2 co-folding with cathepsin B (full panel)

Linker	type	ipTM	pLDDT	Kd (nM)
Maleimide/Val-Ala-PABC	designed	0.80	89.7	17.8
Oxyamine/Val-Ala-PABC	designed	0.82	88.2	623.4
NHS_ester/Val-Ala-PABC	designed	0.70	85.5	21.4
Bromoacetamide/Val-Ala-PABC	designed	0.72	84.4	8.1
Maleimide/Val-Cit-PABC	designed	0.81	84.9	7.5
Disulfide/Val-Ala-PABC	designed	0.82	86.8	11.3
mc-Val-Cit-PABC	commercial	0.87	87.4	33.0
mc-Val-Ala-PABC	commercial	0.83	87.8	129.6
maleimidocaproyl (mc)	commercial	0.86	91.5	90176.0
AcBut hydrazone	commercial	0.77	89.5	21150.8